

SUPPLEMENTAL PROBLEM SET

Show that the value of the chooser option as given by either Equation A.61 or A.62 is less than the value of the straddle.

Hint: You may find it useful to make use of your results from Problem 12 in this problem set.

e) Consider a chooser option on a non-dividend-paying stock with a final maturity of one year. Assume the stock price is lognormally distributed. In three months, the owner determines whether the option is a call or a put with a strike price of \$50. The current stock price is \$55, the volatility of the stock is 20% per annum, and the continuously compounded risk-free rate is 6%. What is the value of the chooser today?

Appendix B

A Brief Review of Some Fixed Income Topics

The purpose of this appendix is to provide a very brief review of the concept of duration as a primary measure of risk for bonds. For simplicity, we will consider here only bonds that are assumed to pay one time per year, and in addition we will only consider the valuation and analysis of bonds on their settlement date. For a more general discussion, see, for example, Fabozzi (1993).

B.1 Present Value

Perhaps the most fundamental concept of valuation of fixed income securities is the notion of present value. If a cash flow c is to be received in t periods, then the present value of that cash flow is equal to

$$PV = \frac{c}{[1 + zc(t)]^t}, \quad (\text{B.1})$$

where $zc(t)$ represents an appropriate discount rate, or *zero coupon rate*, for cash flow received at time t . A concept related to a zero coupon rate is a *discount factor*. The discount factor for cash flow received at time t , denoted $DF(0, t)$, is given by

$$DF(0, t) = \frac{1}{[1 + zc(t)]^t}. \quad (\text{B.2})$$

It is important to note the dependence of the particular zero coupon rate on the time at which the cash is received. Cash flows received at different times

will be discounted using different zero coupon rates. For example, suppose we have an n -year bond that pays an annual coupon equal to c , then its price today, P , is given by

$$P = \frac{c}{[1 + zc(1)]^1} + \frac{c}{[1 + zc(2)]^2} + \cdots + \frac{c + F}{[1 + zc(n)]^n}, \quad (\text{B.3})$$

where F is the face value of the bond. Given the price of a bond, P , we can define the *yield to maturity*, y , of the bond as that number that solves the following equation:

$$P = \frac{c}{(1 + y)^1} + \frac{c}{(1 + y)^2} + \cdots + \frac{c + F}{(1 + y)^n}. \quad (\text{B.4})$$

Note that the yield to maturity is the yield used to discount each cash flow of the bond, regardless of when it is received. It is important to note that yield to maturity is a market convention; it is much easier to analyze a bond and make a comparison of bonds on the basis of a single number, yield to maturity, than it is to discuss a whole sequence of zero coupon rates. However, yield to maturity has no true economic meaning. Individual cash flows should rightfully be discounted at their individual zero coupon rates, not the artificial construct known as yield to maturity.

B.2 Duration

Duration is the primary metric used in fixed income to measure the price sensitivity of a bond to changes in interest rates. We will look at three different measures of duration. Macaulay duration is a weighted average time until the cash flows of a bond are received. This measure is of limited practical use. Modified duration is often used to measure price sensitivity when the cash flows of a bond are known with certainty. When the cash flows of a bond are uncertain due to options that have been embedded in the bond, a popular measure of interest rate sensitivity used is referred to as effective duration.

B.2.1 Macaulay Duration

Define

$$\text{PVCF}_i = \frac{\text{CF}_i}{(1 + y)^i},$$

where CF_i denotes the cash flow of the bond at time i . Note that

$$P = \sum_{i=1}^n \text{PVCF}_i.$$

Macaulay duration is defined by

$$\frac{\sum_{i=1}^n i \times \text{PVCF}_i}{P}.$$

Macaulay duration is equal to the weighted average time to receiving cash flow of the bond, where the weights are equal to present value of each cash flow (computed using its yield to maturity y) as a percentage of the bond's price. Macaulay duration is of limited use in the analysis of bonds. There is a related measure, however, that is widely used.

B.2.2 Modified Duration

The *modified duration* of a bond, D_m , is defined by

$$D_m = \frac{\text{Macaulay duration}}{1 + y}.$$

The modified duration of a bond, D_m , has the property that

$$dP = -D_m \times P \times dy, \quad (\text{B.5})$$

where dy represents a small change in the yield to maturity of the bond and dP represents the change in the price of the bond. Thus, given the modified duration of a bond, its price, and a small change in yield, Equation B.5 gives us a way to compute the change in the price of the bond. It is very important to note that the modified duration of a bond is not a constant but rather changes as a function of the yield to maturity. As a result, the formula in Equation B.5 is accurate only for a small change in yield. The difference between the actual change in a bond's price and the change in price as given in Equation B.5 is measured by the *convexity* of the bond. The convexity can be quantified, and the interested reader is referred to Fabozzi (1993).

Example

Figure B.1 shows the price of a 30-year bond with an annual coupon of 6% as a function of its yield to maturity. The price is computed using Equation B.4 and is represented by the curve in the figure. Given an initial yield to maturity of 6%, the modified duration of the bond has been calculated and is equal to 14.59. For a given change in yield, Equation B.5 is used to estimate the new price of the bond. These estimates are represented by the straight line in the graph. Note that the difference between the actual price of the bond for a given change in yield and the estimate of its price based on its modified duration is a positive number. The convexity of the bond quantifies this difference. $\square$

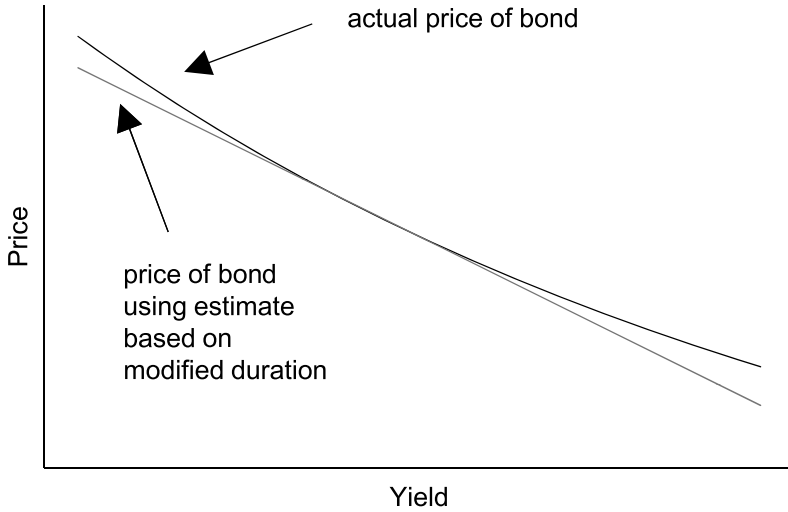


Figure B.1: A Bond with Positive Convexity and Its Price Estimates Based on Its Modified Duration

B.2.3 Effective Duration

The bonds that we have considered thus far have known cash flows. Some bonds that we consider have embedded options, and as a result the coupons (or principal) paid by such bonds are not certain but depend on those options. It is often desirable to come up with a measure of these bonds' sensitivities to changes in rates. The use of modified duration is inappropriate since modified duration assumes that the coupons are known with certainty.

For bonds with embedded options, an often used metric for price sensitivity is known as *effective duration*. The effective duration of the bond is given by

$$\frac{P_{-\Delta x} - P_{+\Delta x}}{2 \times \Delta x \times P_0}, \quad (\text{B.6})$$

where $P_{-\Delta x}$ and $P_{+\Delta x}$ are the values of the bond when the yield curve is shifted down or up by an amount Δx , respectively, and P_0 represents the value of the bond when the yield curve is not shifted. Notice that the values $P_{-\Delta x}$ and $P_{+\Delta x}$ incorporate changes to the value of the bond due to any embedded options. Furthermore, note that this measure of interest rate sensitivity does not rely on a single yield to maturity concept but rather on a parallel shift in the underlying yield curve up or down.

Appendix C

A Closer Look at Day Count and Payment Conventions in Swaps

The standard settlement date, or effective date, on USD *spot starting swaps* is two New York and London Business Days after the trade date, often denoted “ $T + 2$.”¹ Unless otherwise indicated, the standard convention is that the fixed side pays semiannually on a 30/360 day count basis and the floating side is 3-month LIBOR that pays quarterly on an Act/360 basis.

Payments are never made on “bad days,” defined as weekends or holidays in New York or London during which time banks or foreign exchange markets are closed. If a payment were to fall on a bad day, the actual payment date will depend on whether payments are made on a *preceding*, *following*, or *modified following* basis. If payments are made on a preceding day basis, then if a payment were due on a bad day, it would instead be paid on the first preceding good business day. If payments are made on a following day basis, then if a payment were due on a bad day, it would instead be paid on the next good business day. Finally, if payments are made on a modified following basis, then if a payment were due on a bad day, it would instead be paid on the next good business day unless that next good business day happens to fall in the subsequent month, in which case we revert to making the payment on the first preceding good business day.

Preceding, following, or modified following answer *when* payments should occur. But if a payment date is moved in accordance with one of these conventions, should the payment amount be adjusted as well? Exactly *how much* should be paid on a payment date is determined by whether payments

¹ *Forward starting swaps* have effective dates that are beyond $T + 2$.

Fixed Side:			Floating Side:		
Accrual Start	Accrual End	Days	Accrual Start	Accrual End	Days
Wed, Apr 30, 03	Thu, Oct 30, 03	180	Wed, Apr 30, 03	Wed, Jul 30, 03	91
Thu, Oct 30, 03	Fri, Apr 30, 04	180	Wed, Jul 30, 03	Thu, Oct 30, 03	92
Fri, Apr 30, 04	Fri, Oct 29, 04	179	Thu, Oct 30, 03	Fri, Jan 30, 04	92
Fri, Oct 29, 04	Fri, Apr 29, 05	180	Fri, Jan 30, 04	Fri, Apr 30, 04	91
			Fri, Apr 30, 04	Fri, Jul 30, 04	91
	Sum	719	Fri, Jul 30, 04	Fri, Oct 29, 04	91
			Fri, Oct 29, 04	Mon, Jan 31, 05	94
			Mon, Jan 31, 05	Fri, Apr 29, 05	88
				Sum	730

Table C.1: Swap Payment Dates in 2-Year Spot Starting Swap Executed on April 28, 2003, Assuming Both the Fixed and Floating Sides Pay Modified Following with Adjustment

are made on an *adjusted* or *unadjusted* basis. If a payment date falls on a bad day, then the actual date on which the payment occurs is changed, and this results in changing the number of days in the period. If payments are made with adjustment, then the number of accrual days in the period will change to correspond with the change in the payment date. If payments are made with no adjustment, then the number of accrual days in the period will not change despite the change in the payment date.

The standard in the market is that swap payments are made modified following with adjustment on both the fixed and floating sides. Let's take a look at a 2-year swap executed on April 28, 2003, which is a Monday. The effective date of the swap, following the standard $T + 2$ rule, is Wednesday, April 30, 2003. The actual payment dates of the swap are shown in Table C.1.

Note that there are a total of 719 days that accrue on the fixed side and 730 accrual days on the floating side. Although this may seem “unfair” to the floating rate payer who has to pay out accruals for an additional 11 days, the fixed rate takes that into account, and there is no arbitrage opportunity here. The fixed rate is set such that the present value of the fixed cash flows equals the present value of the floating cash flows and incorporates the disparity in the number of accrual days on each side of the swap.

The fixed side pays semiannually, and if there were no bad days, this would imply payments would occur each October 30 and April 30. Note, however, that one of the fixed payments occurs on October 29, 2004. The

Fixed Side:			Floating Side:		
Accrual Start	Accrual End	Days	Accrual Start	Accrual End	Days
Wed, Apr 30, 03	Thu, Oct 30, 03	180	Wed, Apr 30, 03	Wed, Jul 30, 03	91
Thu, Oct 30, 03	Fri, Apr 30, 04	180	Wed, Jul 30, 03	Thu, Oct 30, 03	92
Fri, Apr 30, 04	Mon, Nov 1, 04	181	Thu, Oct 30, 03	Fri, Jan 30, 04	92
Mon, Nov 1, 04	Tues, May 3, 05	182	Fri, Jan 30, 04	Fri, Apr 30, 04	91
			Fri, Apr 30, 04	Fri, Jul 30, 04	91
	Sum	723	Fri, Jul 30, 04	Fri, Oct 29, 04	91
			Fri, Oct 29, 04	Mon, Jan 31, 05	94
			Mon, Jan 31, 05	Fri, Apr 29, 05	88
				Sum	730

Table C.2: Swap Payment Dates in 2-Year Spot Starting Swap Executed April 28, 2003 Assuming the Fixed Side Pays Following with Adjustment and Floating Side Pays Modified Following with Adjustment

reason for this is that October 30, 2004 is a Saturday. The next good business day is November 1, 2004, but inasmuch as that takes us into the next month, under modified following we revert back to the first preceding good business day, which is Friday, October 29. Since payments are adjusted, the fixed rate accrues for only 179 days this period, the number of 30/360 days between April 30, 2004 and October 29, 2004. The final period on the fixed side faces a similar plight: April 30, 2005 is a Saturday, and since the next good business day is in May, we revert back to Friday, April 29, 2005 for the last payment date in the swap under modified following. Note that the number of bond days in this period from October 29, 2004 to April 29, 2005 is 180. If the fixed side were unadjusted, each semiannual period would accrue exactly 180 days despite the change in actual payment dates.

The same conventions apply to the floating leg of the swap and affect the quarterly periods ending in October 2004 and April 2005 similarly. In addition, the quarterly period ending on January 30, 2005 is a Sunday, and thus the payment is made on Monday, January 31, 2005 under the modified following convention.

Now let's take a look at a 2-year swap that is also executed on Monday, April 28, 2003 and also has an effective date of April 30, 2003, but assume that payments on the fixed side are made on a following basis with adjustment. Assume the floating payments are still made on a modified following basis with adjustment. The actual payment dates of this swap are shown in Table C.2.

The floating payments in this swap are identical to the ones in Table C.1, but the fixed payments differ somewhat. In the third semiannual period, the fixed payment is now made on Monday, November 1, 2004; the scheduled payment date of October 30, 2004 was a bad day, and under the following convention we roll to the next good business day. Since payments are still made with adjustment, this means that we accrue the swap rate for the total number of bond days between April 30, 2004 and November 1, 2004 which is 181. We also need to be careful with the last semiannual period in the swap. April 30, 2005 is a Saturday, and thus we need to roll forward to the next good business day. Sunday is obviously out, but we need be careful: Monday, May 2, 2005 is May Day, a London holiday on which banks are closed. Thus the next good business day is Tuesday, May 3, 2005. Since payments are made with adjustment, the swap rate is accrued for the number of bond days between November 1, 2004 and May 3, 2005 which is 182. If the fixed side were accruing on an unadjusted basis, each period would accrue for exactly 180 days despite the change in payment dates.

The upshot here is that both 2-year swaps (i.e., the one in which the fixed side pays modified following with adjustment and the one in which the fixed side pays following with adjustment) are not the same. Both have of course the same exact cash flows on the floating side, but their fixed sides differ due to the different payment conventions employed. The first swap has a total of 719 bond days accruing on the fixed side, whereas the second has a total of 723 days. The fixed rates in these two swaps cannot be the same. If the fixed rates were the same, we would always be better off paying in the first one and receiving in the second one at the same fixed rate and accrue the additional 4 days. Which swap should have the higher rate?

As of 3:00 p.m. EST on April 28, 2003, the fixed rate in the first of the 2-year swaps in which both sides pay modified following with adjustment was 1.81643%. At the very same time, the fixed rate in the second of the two swaps in which the fixed side pays following with adjustment was 1.80671%, a difference of about 0.97 bps. This is a fairly substantial difference: let's say we entered into \$1bn notional of this 2-year swap and agreed to pay fixed on a following basis, but paid at the wrong rate of 1.81643%. We would be paying a rate that is too high and would have lost around \$190,000 on a mark-to-market basis for our mistake.²

Back in the 1990s, some proprietary swap models failed to take these differences into account. At the time, there were some traders in the market who would try to "pick off" (i.e., take advantage of) other traders whose models did not account for these adjustments properly. For example, suppose

²To figure out how much we would lose, put $PMT = 0.97/2$ bp, $n = 4$, and $i = 1.80671\%/2$ and get $PV = 1.9$ bps upfront, or \$190,000 on a notional of \$1bn.

we were aware of the differences in how these two swaps should be priced but our counterparty was not. We could enter into a \$1bn 2-year swap with them in which we receive fixed using standard market conventions of modified following on both sides. Suppose we receive fixed in this swap at 1.81643%. After the deal had been executed, we could contact our counterparty about our (seemingly innocuous) request to change the fixed side of the swap to following with adjustment, instead of modified following with adjustment, but keep the fixed rate the same. Our counterparty might very well agree to the change, since his model fails to price the difference correctly. In reality, when changing the fixed leg to following with adjustment, the fixed rate in the swap should decline to 1.80671%. Thus, we would make nearly \$200,000 by picking off our counterparty for nearly 1 bp in the rate. Of course, such behavior would be considered a huge breach of etiquette by many in the market. That being said, the conventional wisdom in the market is that certain traders attempted to exploit opportunities like this during that time.

It is important to note that significant differences in swap rates as a function of payment conventions are exceedingly rare. Such differences tend to manifest themselves in swaps whose payment dates roll near the end of months, such as in the example considered here. Typically, the difference in a swap by changing the payment conventions is a small rounding error. But swaps that have payment dates near the end of a month should be priced with some extra caution, as hopefully this example demonstrates.